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# Airport Network Design and Implementation

**Zone-Based Segmentation, Customer Help Desk and ACL Security**

**Case Study Report**

# 1. Introduction

This case study presents the design and implementation of a network for an airport, built and tested in Cisco Packet Tracer. The network connects the airport's departments (Administration, IT, Check-in, Security, Baggage, Operations and a Customer Help Desk), its servers, and staff and passenger Wi-Fi, through a core router with an edge router towards the outside network.

Departments are grouped into logical zones, each with its own switch, and separated from one another using VLANs, inter-VLAN routing and Access Control Lists (ACLs), so that every department can reach only the resources it needs.

# 2. Aim

To design and implement a logically and physically segmented, secure airport network in Cisco Packet Tracer, using VLANs, inter-VLAN routing, DHCP and ACLs, and to verify that each department can reach only the resources it should.

# 3. Objectives

- Group departments into logical zones, each connected to its own switch, so the topology matches the organisation.
- Provide every department, including the Customer Help Desk, with its own VLAN, subnet and DHCP pool.
- Provide automatic addressing (DHCP) and inter-VLAN routing for all departments.
- Restrict access between departments using extended ACLs, while keeping shared services (DNS, web) available.
- Test and document the results, including blocked and allowed traffic.

# 4. Network Design

## 4.1 Scenario

The airport network is split into five zones: Administration & IT, Operations & Security, Services, Wireless, and the Customer Help Desk. Each zone has one switch, and the switches connect to the core router R1-Core, which performs inter-VLAN routing (router-on-a-stick) and also runs the DHCP service. R1-Core connects to R2-Edge, which represents the edge towards the outside network.

Check-in, Security, Baggage and Operations PCs connect to SW2-Operations, Admin and IT PCs to SW1-Admin, the servers to SW3-Services, the access points to SW4-Wireless, and the help desk PCs to SW-HelpDesk (Figure 1).

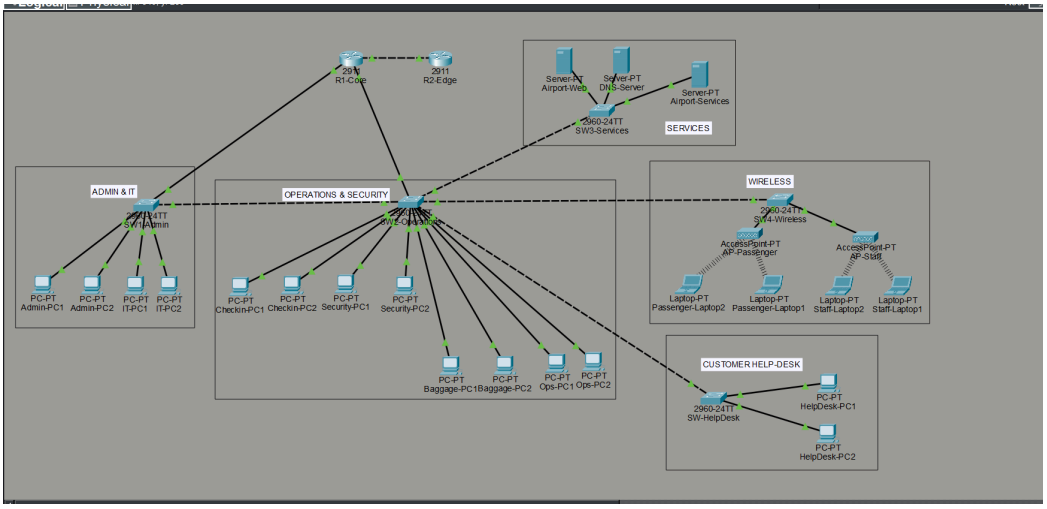



*Figure 1: Airport network topology with logical zones*

### 4.2 Devices Used

| Device | Model | Qty | Role |
|---|---|---|---|
| R1-Core, R2-Edge | Cisco 2911 router | 2 | R1-Core: inter-VLAN routing, DHCP, ACLs. R2-Edge: edge router. |
| SW1-Admin, SW2-Operations, SW3-Services, SW4-Wireless, SW-HelpDesk | Cisco 2960-24TT switch | 5 | One switch per zone. |
| AP-Staff, AP-Passenger | AccessPoint-PT | 2 | Wireless access for staff and passengers. |
| Department PCs | PC-PT | 14 | Admin 2, IT 2, Check-in 2, Security 2, Baggage 2, Operations 2, Help Desk 2. |
| Staff and passenger laptops | Laptop-PT | 4 | Connect wirelessly to the access points. |
| Airport-Web, DNS-Server, Airport-Services | Server-PT | 3 | Web portal, name resolution and airport services. |

| Zone | Switch | Departments (VLAN) | End devices |
|---|---|---|---|
| Administration & IT | SW1-Admin | Admin (10), IT (60) | 4 PCs |
| Operations & Security | SW2-Operations | Check-in (20), Security (30), Baggage (40), Operations (50) | 8 PCs |
| Services | SW3-Services | Servers (90) | 3 servers |
| Wireless | SW4-Wireless | Staff Wi-Fi (70), Passenger Wi-Fi (80) | 2 APs, 4 laptops |
| Customer Help Desk | SW-HelpDesk | Help Desk (100) | 2 PCs |

## 5. IP Configuration

Each VLAN has its own /24 subnet of the form 192.168.<VLAN>.0/24, with the router sub-interface as the default gateway (.1). R1-Core connects to R2-Edge over the 10.0.0.0 link (R1-Core G0/0 = 10.0.0.1). VLANs 10 and 20 are routed on the G0/1 trunk (towards SW1-Admin); VLANs 30 to 100 are routed on the G0/2 trunk (towards SW2-Operations).

| VLAN | Department | Subnet | Gateway | Addressing |
|---|---|---|---|---|
| 10 | Administration | 192.168.10.0/24 | 192.168.10.1 | DHCP |
| 20 | Check-in | 192.168.20.0/24 | 192.168.20.1 | DHCP |
| 30 | Security | 192.168.30.0/24 | 192.168.30.1 | DHCP |
| 40 | Baggage | 192.168.40.0/24 | 192.168.40.1 | DHCP |
| 50 | Operations | 192.168.50.0/24 | 192.168.50.1 | DHCP |
| 60 | IT | 192.168.60.0/24 | 192.168.60.1 | DHCP |
| 70 | Staff Wi-Fi | 192.168.70.0/24 | 192.168.70.1 | DHCP |
| 80 | Passenger Wi-Fi | 192.168.80.0/24 | 192.168.80.1 | DHCP |
| 90 | Servers | 192.168.90.0/24 | 192.168.90.1 | Static (DNS: .10) |
| 100 | Customer Help Desk | 192.168.100.0/24 | 192.168.100.1 | DHCP |

DHCP pools run on R1-Core, one per VLAN. The first 10 addresses of each subnet are excluded from the pool for infrastructure use, and every pool hands out the DNS server address 192.168.90.10.

## 6. Implementation Steps

1. **Planned the zones and addressing:** one VLAN and one /24 subnet per department (Section 5), and one switch per zone.

2. **Built the topology:** R1-Core connects to R2-Edge (G0/0), SW1-Admin (G0/1) and SW2-Operations (G0/2). SW3-Services, SW4-Wireless and SW-HelpDesk connect to SW2-Operations, and every end device connects to the switch of its own zone.
3. **Configured VLANs and trunks on the switches:** created the VLANs, assigned the access ports and set the links between switches as 802.1Q trunks. Example configuration:

```
! SW2-Operations
vlan 20
 name Checkin
vlan 50
 name OPERATIONS
vlan 100
 name HELPDESK
interface range fastEthernet0/21 - 22
 switchport mode access
 switchport access vlan 20
interface range fastEthernet0/19 - 20
 switchport mode access
 switchport access vlan 50
interface fastEthernet0/23
 switchport mode trunk
! SW1-Admin
vlan 60
 name IT
interface range fastEthernet0/21 - 22
 switchport mode access
 switchport access vlan 60
! SW-HelpDesk
vlan 100
 name HELPDESK
interface fastEthernet0/24
 switchport mode trunk
interface range fastEthernet0/1 - 3
 switchport mode access
 switchport access vlan 100
```

4. **Configured inter-VLAN routing on R1-Core** with one 802.1Q sub-interface per VLAN (VLANs 10 and 20 on G0/1, VLANs 30 to 100 on G0/2). Example:

```
interface gigabitEthernet0/1.10
 encapsulation dot1Q 10
 ip address 192.168.10.1 255.255.255.0
interface gigabitEthernet0/2.100
 encapsulation dot1Q 100
 ip address 192.168.100.1 255.255.255.0
! same pattern for VLANs 20, 30, 40, 50, 60, 70, 80 and 90
```

5. **Configured DHCP on R1-Core:** one pool per VLAN with the gateway and DNS server. Example:

```
ip dhcp excluded-address 192.168.100.1 192.168.100.10
ip dhcp pool VLAN100-HELPDESK
 network 192.168.100.0 255.255.255.0
 default-router 192.168.100.1
 dns-server 192.168.90.10
```

6. **Set up the servers** with static addresses in 192.168.90.0/24 (DNS server 192.168.90.10) and the airport web page on Airport-Web.
7. **Connected the wireless zone:** staff and passenger laptops connect through AP-Staff and AP-Passenger on SW4-Wireless.
8. Applied two ACLs on R1-Core to control access between departments (Section 8).
9. Tested connectivity, DHCP, the ACLs and the web service (Section 9), and saved the configuration on every device.

# 7. Tools and Technologies

- **Cisco Packet Tracer** for design, configuration and simulation.
- **VLANs and 802.1Q trunking** for logical separation of departments across switches.
- **Router-on-a-stick** (sub-interfaces on R1-Core) for inter-VLAN routing.

- **DHCP, DNS and HTTP** services for automatic addressing, name resolution and the airport web portal.
- **Extended Access Control Lists (ACLs)** for traffic filtering between departments.
- **Wireless access points** for staff and passenger Wi-Fi; ping (ICMP) and the web browser for verification.

## 8. Network Security Features

Security is built in three layers: (1) VLAN segmentation, so each department has its own broadcast domain and subnet; (2) zone-based physical separation, so each department's devices connect to the switch of its own zone; and (3) extended ACLs on R1-Core that control which departments can talk to each other. Both ACLs are applied inbound on the sub-interface of the VLAN they protect against, and each ends with permit ip any any so that all other traffic keeps working.

| ACL name | Applied on | Blocked | Still allowed |
|---|---|---|---|
| PASSENGER-RESTRICT | G0/2.80 (in) | Passenger Wi-Fi to Admin, Check-in, Security, Baggage, Operations, IT, Staff Wi-Fi and Help Desk subnets | Servers (VLAN 90) such as the web portal and DNS |
| HELPDESK-RESTRICT | G0/2.100 (in) | Help Desk to Admin (VLAN 10) and IT (VLAN 60) | Servers and all other traffic |

```
ip access-list extended PASSENGER-RESTRICT
 deny ip 192.168.80.0 0.0.0.255 192.168.10.0 0.0.0.255
 ... (same deny line for 20, 30, 40, 50, 60, 70 and 100)
 permit ip any any
interface gigabitEthernet0/2.80
 ip access-group PASSENGER-RESTRICT in

ip access-list extended HELPDESK-RESTRICT
 deny ip 192.168.100.0 0.0.0.255 192.168.10.0 0.0.0.255
 deny ip 192.168.100.0 0.0.0.255 192.168.60.0 0.0.0.255
 permit ip any any
interface gigabitEthernet0/2.100
 ip access-group HELPDESK-RESTRICT in
```

## 9. Testing and Results

| # | Test | Expected | Result |
|---|---|---|---|
| 1 | Checkin-PC1 and PC2 get a VLAN 20 address by DHCP and ping 192.168.20.1 | Replies | Pass (4/4 replies) |
| 2 | Ops-PC1 (192.168.50.11) and Ops-PC2 (192.168.50.12) ping 192.168.50.1 | Replies | Pass (4/4 replies) |
| 3 | IT-PC1 and IT-PC2 get a VLAN 60 address by DHCP and ping 192.168.60.1 | Replies | Pass (4/4 replies) |
| 4 | HelpDesk-PC1 (192.168.100.11) and PC2 (192.168.100.12) get DHCP addresses and ping 192.168.100.1 | Replies | Pass (all replies received) |
| 5 | HelpDesk-PC1 pings the DNS server 192.168.90.10 | Replies | Pass (all replies received) |
| 6 | Passenger-Laptop1 (192.168.80.12) pings Admin gateway 192.168.10.1 | Blocked by ACL | Pass (Destination host unreachable) |
| 7 | HelpDesk-PC1 pings Admin gateway 192.168.10.1 | Blocked by ACL | Pass (Destination host unreachable) |
| 8 | Admin-PC1 opens the airport web page | Page loads | Pass |
| 9 | Passenger-Laptop1 opens the airport web page | Page loads (servers allowed) | Pass |

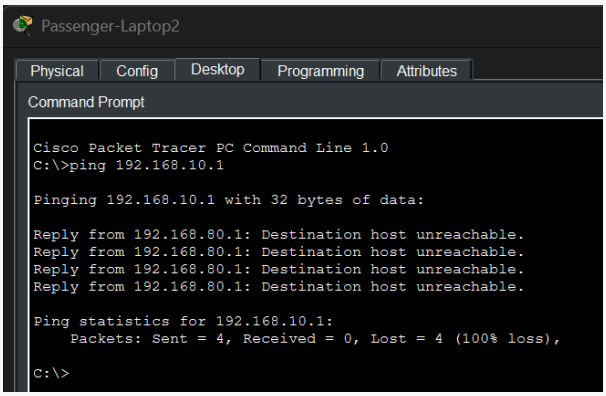



*Figure 2: Passenger Wi-Fi to Admin, blocked*

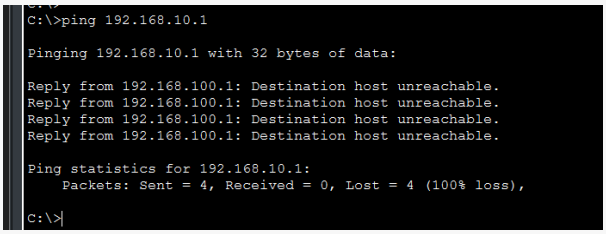



*Figure 3: Help Desk to Admin, blocked*

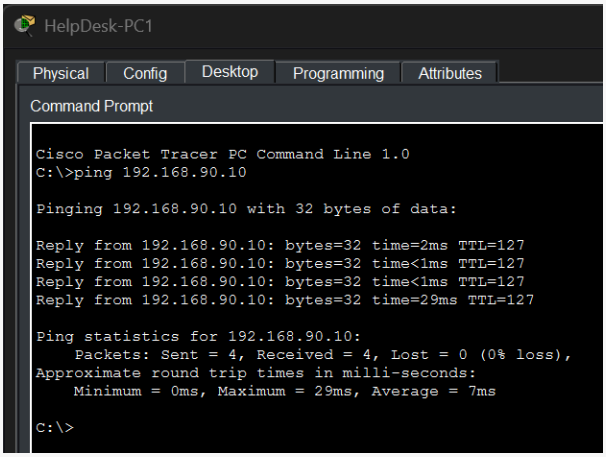



*Figure 4: Help Desk to server, allowed*

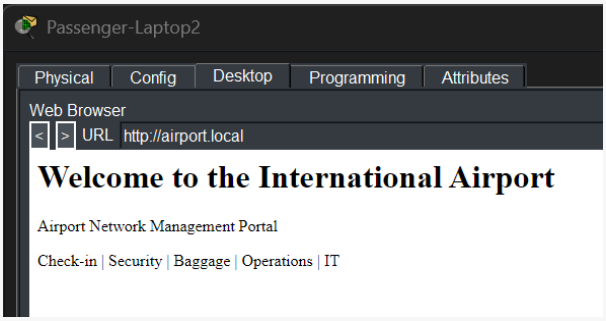



*Figure 5: Web page working*

## 10. Conclusion

The airport network follows a clear zone-based structure: every department connects to the switch of its own zone, the servers are isolated on their own switch, and the Customer Help Desk is a separate department with its own VLAN and subnet. DHCP, inter-VLAN routing and the airport web service work across all zones. The two ACLs on R1-Core prevent Passenger Wi-Fi and the Help Desk from reaching sensitive internal departments, while still allowing them to use the shared servers, and the tests confirmed both the blocked and the allowed traffic.

## 11. Future Enhancements

- Add a core/distribution switch with redundant uplinks and EtherChannel, so the zone switches do not all depend on SW2-Operations.
- Move the VLAN 20 (Check-in) sub-interface to the G0/2 trunk, so Check-in traffic no longer crosses the SW1-to-SW2 link to reach its gateway.
- Harden devices: port security, shutting unused ports, SSH, enable secret and encrypted passwords on routers and switches.
- Enable WPA2 on both access points, and add ACLs for Staff Wi-Fi and Security.
- Add DHCP snooping, a firewall at the edge, and more end devices and services (email, FTP) as the airport grows.

## 12. References


1. Cisco Networking Academy, Cisco Packet Tracer documentation.
2. Cisco, IOS Security Configuration Guide: Access Control Lists.
3. IEEE 802.1Q, Virtual Bridged Local Area Networks (VLAN tagging).
4. R. Droms, RFC 2131: Dynamic Host Configuration Protocol; P. Mockapetris, RFC 1035: Domain Names.
5. Behrouz A. Forouzan, Data Communications and Networking, McGraw-Hill.